Ch 6.3: PCR

Lecture 20 - CMSE 381

Michigan State University
::
Dept of Computational Mathematics, Science & Engineering

Wed, Mar 11, 2026

Announcements

Last time:

- PCA

This lecture:

- PCR

Announcements:

- HW4 due Sunday 3/15.
- Exam #2 on Monday 3/16!
  - Bring 8.5x11 sheet of paper
  - Handwritten both sides
  - Anything you want on it, but must be your work
  - Write your name and group number
  - You will turn it in
  - Non-internet calculator
  - questions about project

11	F	2/6	Multiple Logistic Regression / Multinomial Logistic Regression	4.3.4-5	HW #2 Due Mon 2/9	
	M	2/9	***Project Day & Review***			
	W	2/11	**Midterm #1**			
12	F	2/13	**Class not held**			
13	M	2/16	Leave one out and k-fold CV	5.1.1-3		
14	W	2/18	More k-fold CV	5.1.4-5		Q5
15	F	2/20	k-fold CV for classification	5.1.5		
16	M	2/23	Subset selection	6.1		
17	W	2/25	Shrinkage: Ridge	6.2.1		
18	F	2/27	Shrinkage: Lasso	6.2.2		
	M	3/2	Spring Break			
	W	3/4	Spring Break			
	F	3/6	Spring Break		HW #3 Due Sun 3/8	
19	M	3/9	PCA	6.3		
20	W	3/11	PCR	6.3		Q6

Section 1

Previously...

Shrinkage

Find β to minimize

$$RSS = \sum_{i=1}^{n} \left(y_i - \beta_0 - \sum_{j=1}^{p} \beta_j x_{ij} \right)^2$$

subject to:

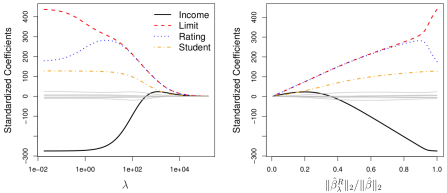
Least Squares:
No constraints

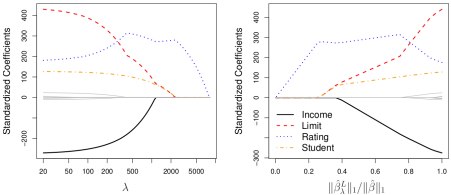
Ridge:

$$\sum_{j=1}^{p} \beta_j^2 \leq s$$

The Lasso:

$$\sum_{j=1}^{p} |\beta_j| \leq s$$





Also, find best choice of λ or s using CV

Linear transformation of predictors

Original Predictors:

$$X_1, \cdots, X_p$$

New Predictors:

$$Z_1, \cdots, Z_M$$

$$Z_m = \sum_{j=1}^{p} \varphi_{jm} X_j$$

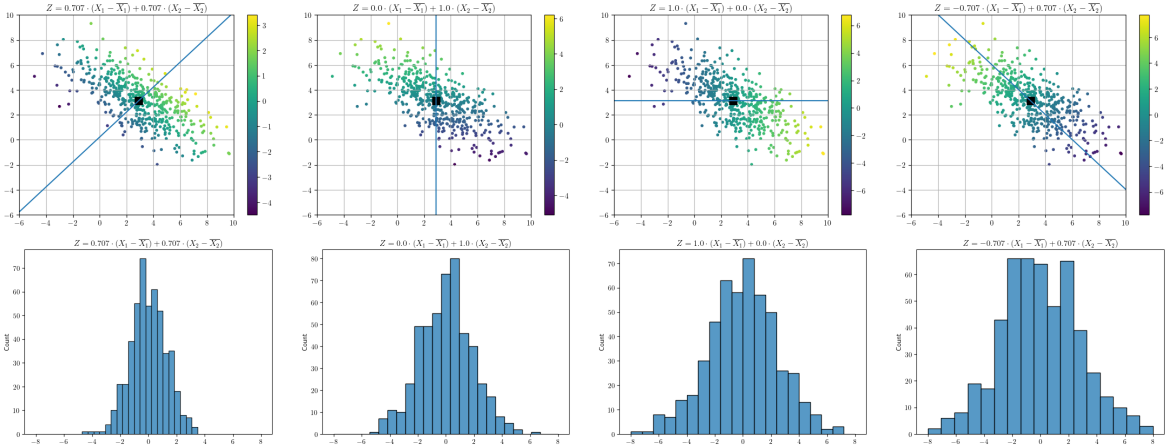
The goal:

- Find good φ's (PCA)
- Fit regression model on Z_i's using least squares (PLS)

$$y_i = \theta_0 + \sum_{m=1}^{M} \theta_m z_{im} + \varepsilon_i$$

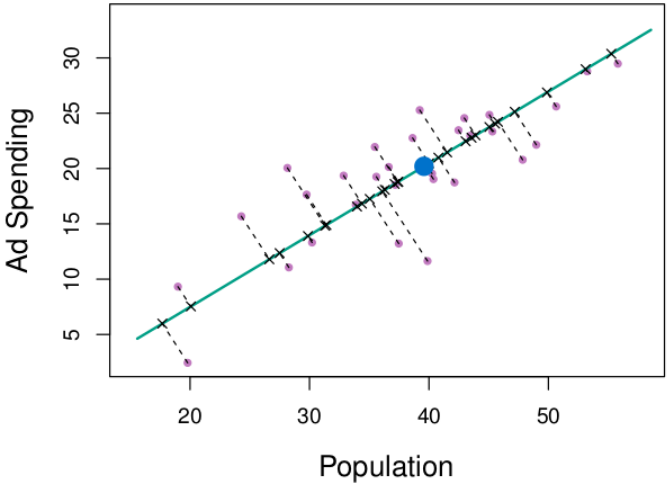
- Hope that lower dimensions means less overfitting

PCA - First PC



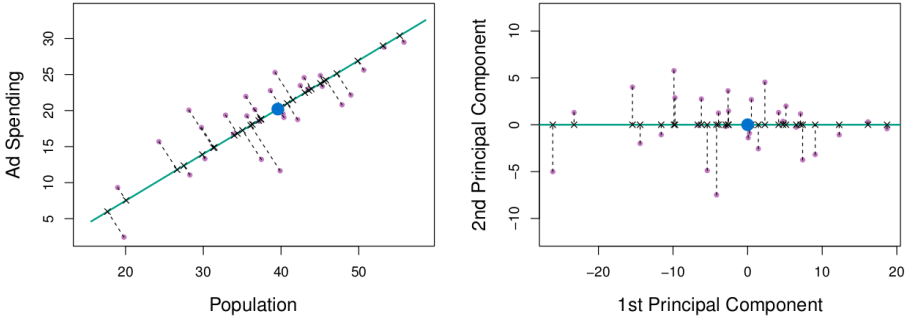
higher variance of projection on the further right pic

Projection onto first PC



$$Z_1 = 0.839 \cdot (\mathtt{pop} - \overline{\mathtt{pop}}) + 0.544 \cdot (\mathtt{ad} - \overline{\mathtt{ad}})$$

Drawing points in PC space



What will you learn from this lecture?

- Why do you want to use the PCs in regression models?
  - What assumptions do you have to make for it to be a good idea to use **principal component regression (PCR)**?
  - Or conversely, what is a typical bad scenario to use PCR?
- How do you implement PCR in Python?
- How do you interpret the model coefficients when using PCR?
- How do you choose the number of PC to use in PCR?
  - Given a figure of, e.g., cross-validation score as a function of the number of PCs, you should be able to choose appropriately and provide rationales in terms of bias-variance tradeoff.
  - You should be able to generate such figures in Python.
- What is the relationship/differences between PCR and feature selection and regularization methods that you learned in this part of the course?

Section 2

Principal Components Regression

So you've found your PCA coefficients

Now what?

- Do linear regression on $Z_1, \cdots, Z_M$
- Book calls this "principal components regression "
- Linear model is $y = \theta_0 + \theta_1 Z_1 + \cdots + \theta_M Z_M$

What are we assuming?

- The directions in which $X_1, \cdots, X_p$ have the most variation are the directions that are associated with Y
- If assumption holds, then fitting on Z_i better than fitting on X_i since fewer variables lessens chance of overfitting

Interpretation of PCR coefficients

Original Predictors:

$$X_1, \cdots, X_p$$

New Predictors:

$$Z_1, \cdots, Z_M$$

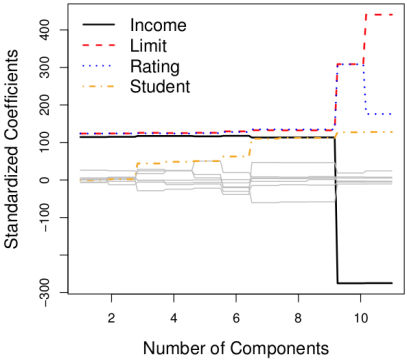
$$Z_m = \sum_{j=1}^{p} \varphi_{jm} X_j$$

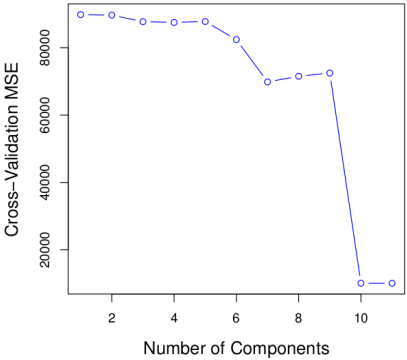
Learned model:

$$y = \theta_0 + \theta_1 Z_1 + \cdots + \theta_M Z_M$$

- Do example showing that coefficients of Z_i can't be used to say anything about X_i's
- $p = 3, M = 2$
- $\varphi = \begin{pmatrix} 1 & -1 \\ 3 & 1/2 \\ 0 & 1/2 \end{pmatrix}$
- Learned model $y = 2Z_1 - Z_2$
- $y = 2(X_1 + 3X_2 + 0X_3) - (-X_1 + \frac{1}{2}X_2 + \frac{1}{2}X_3)$
- $= 3X_1 + 2.5X_2 - 0.5X_3$
- Conclusion: can't directly use coefficients learned to interpret original inputs!
- Each PC is a linear combination of the originals, so can go backwards to figure out whaat the coefficients would have been, see left figure

Picking M





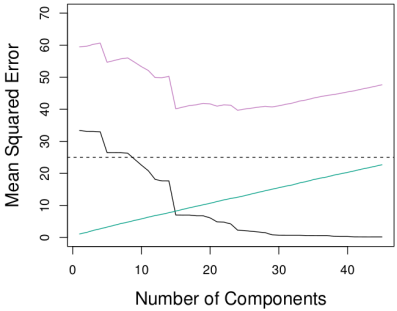
- Credit data set
- Major drop in MSE at 10 compoentns, so barely a savings with the original 11 variables

Do PCR with hitters data

Bias-Variance Trade-off

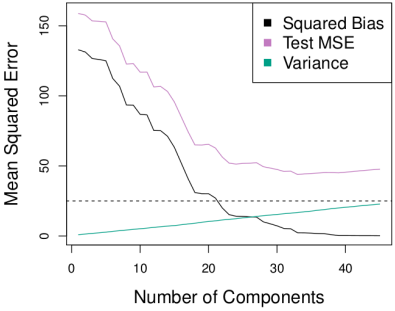
Example with simulated data: $n = 50$ observations of $p = 45$ predictors

Y is a function of **all predictors**



- Typical U shape for Test MSE
- Improve over least square (right most value x=45)

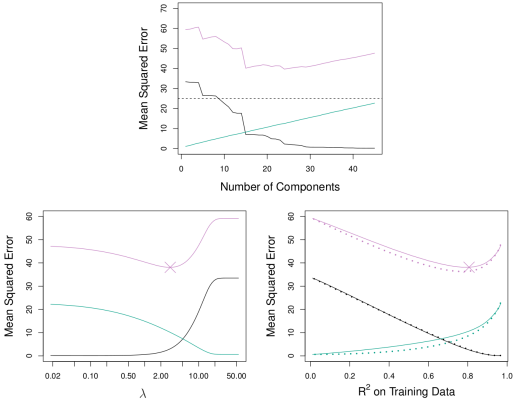
Y is a function of **2 predictors**



- Typical U shape for Test MSE
- (slight) improve over least square (right most value x=45)

Comparison to results on shrinkage

Y is a function of all predictors



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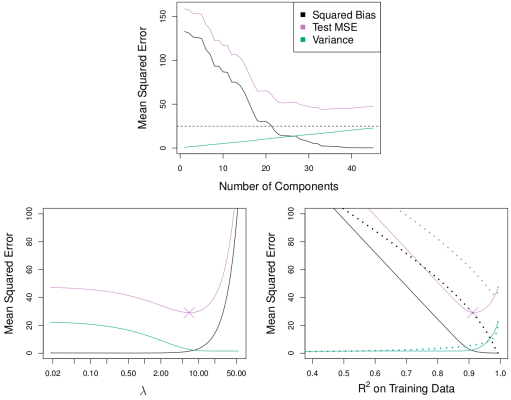


Shrinkage does better

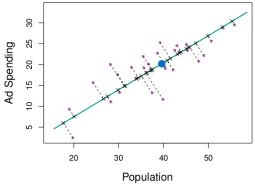
Properties of PCR

- Note better in previous example because lots of PCs needed
- It does better when info contained in the first few PCs
- Number of PCs to use can be picked with CV
- Standardization needed to make sure big data values don't skew the results
- Not a feature selection method because we don't get a subset of features, we get a collection of linear combos of features

TL;DR

PCR

- Unsupervised dimensionality reduction + linear regression
- Choose component Z_1 in the direction of most variance using only X_i's information
- Choose Z_2 and beyond by the same method after "getting rid" of info in the directions already explained



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